

CS 305: Computer Networks

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Lecture 5: Application Layer

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Chapter 2: outline

2.1 principles of network applications

2.2 Web and HTTP

2.3 electronic mail

- SMTP, POP3, IMAP

2.4 DNS

2.5 P2P applications

2.6 video streaming and content distribution networks

2.7 socket programming with UDP and TCP

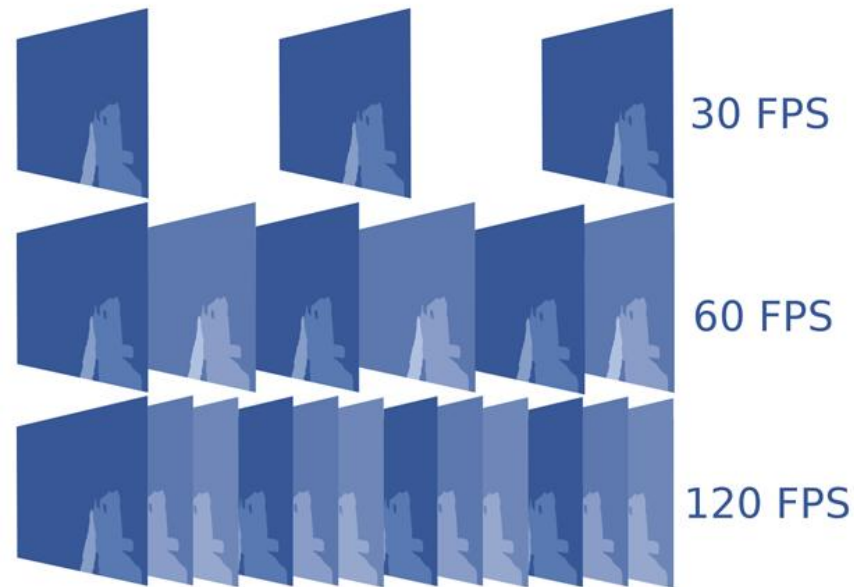
Video Streaming and CDNs: ~~context~~

- Video traffic: major consumer of Internet bandwidth
 - Netflix, YouTube: 37%, 16% of downstream residential ISP traffic
 - ~1B YouTube users, ~75M Netflix users
- Challenge: scale - how to reach ~1B users?
 - single mega-video server won't work (why?)
- Challenge: **heterogeneity**
 - different users have different capabilities (e.g., wired versus mobile; bandwidth rich versus bandwidth poor)
- **Solution:** distributed, application-level infrastructure



Multimedia: video

- Video: sequence of images displayed at constant rate
 - e.g., 24 images/sec
- Digital image: array of pixels
 - each pixel represented by bits

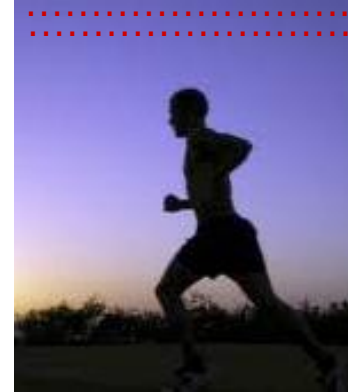


Multimedia: video

Coding (Compression): use redundancy *within* and *between* images to decrease # bits used to encode image

- spatial (within image)
- temporal (from one image to next)

Spatial coding example: instead of sending N values of same color (all purple), send only two values: color value (purple) and number of repeated values (N)



frame i



frame $i+1$

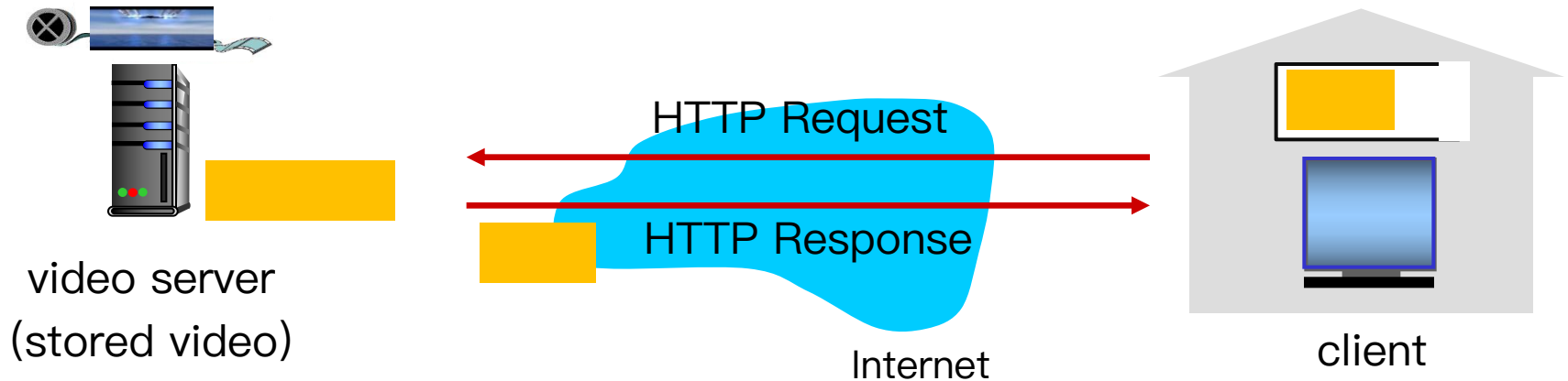
temporal coding example: instead of sending complete frame at $i+1$, send only differences from frame i

Multimedia: video

Type	Video Bitrate, Standard Frame Rate (24, 25, 30)	Video Bitrate, High Frame Rate (48, 50, 60)
2160p (4k)	35-45 Mbps	53-68 Mbps
1440p (2k)	16 Mbps	24 Mbps
1080p	8 Mbps	12 Mbps
720p	5 Mbps	7.5 Mbps
480p	2.5 Mbps	4 Mbps
360p	1 Mbps	1.5 Mbps

- **CBR: (constant bit rate):** video encoding rate fixed
- **VBR: (variable bit rate):** video encoding rate changes as amount of spatial, temporal coding changes

HTTP Streaming



All clients **receive the same encoding of the video:**

- Human users may have different requirements
- Clients may have different available bandwidth, which may be time-varying

How to deal with this?

Streaming multimedia: DASH

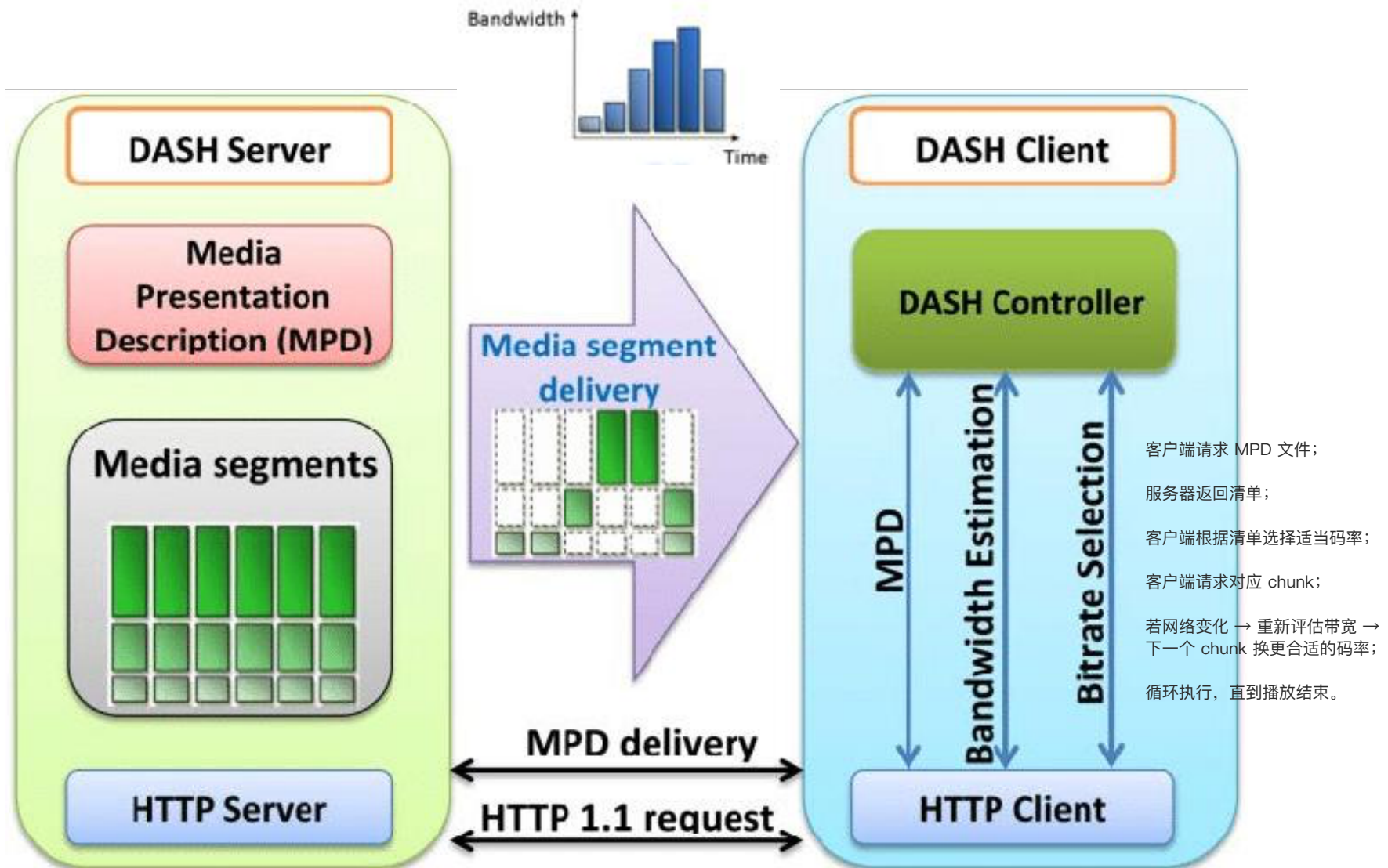
- **DASH: D**ynamic, **A**daptive **S**treaming over **H**TTP
- **Server:**
 - divides video file into **multiple chunks**
 - each chunk **stored, encoded at different rates**
 - **manifest file**: provides URLs for different chunks encoded at different rates
- **Client:**
 - periodically **measures server-to-client bandwidth**
 - consulting manifest, requests one chunk at a time
 - chooses **maximum** coding rate sustainable given current bandwidth 从 Manifest 文件中选择当前带宽能支持的最高码率版本；
 - can choose different coding rates at different points in time (depending on available bandwidth at time)

每个 chunk 请求都是独立的；

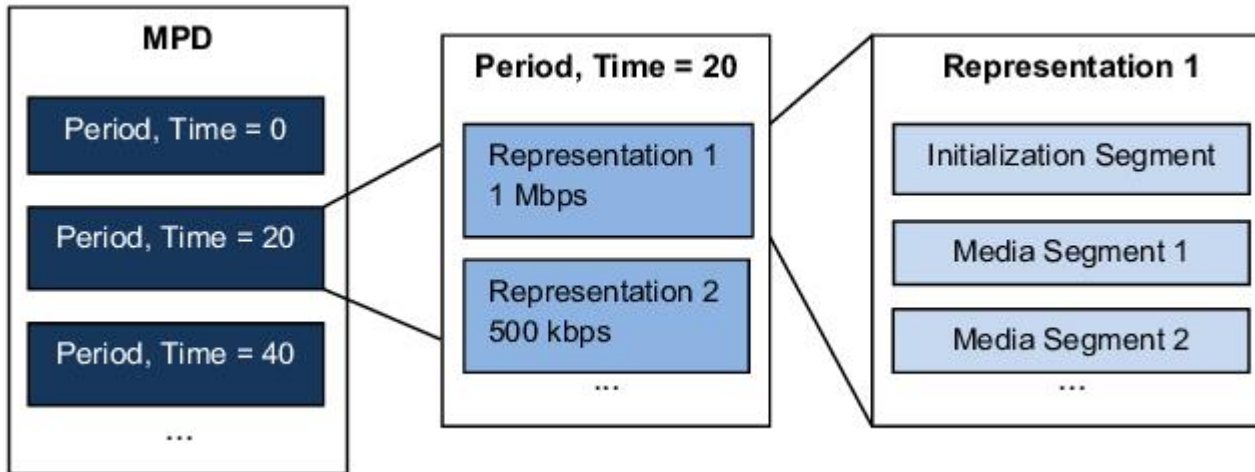
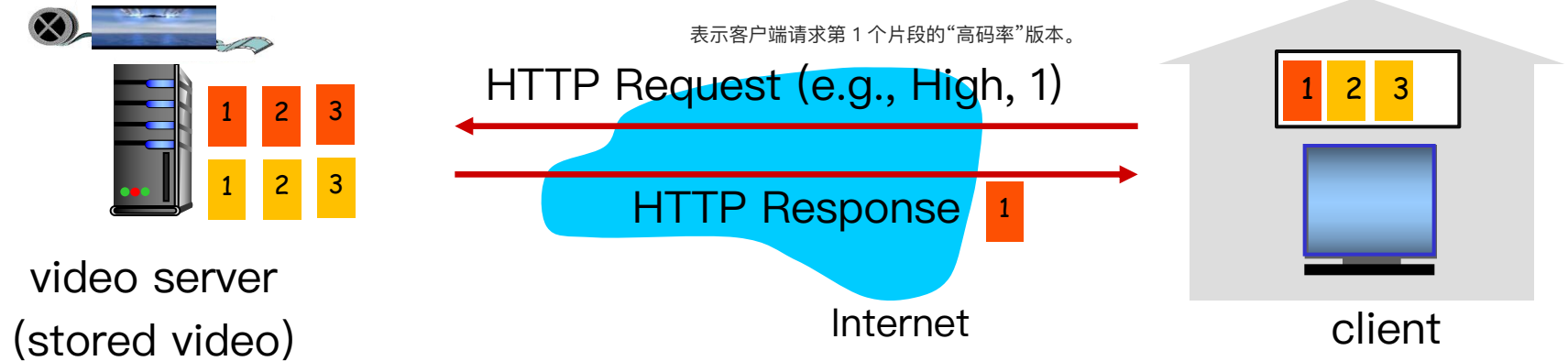
所以客户端可以在每个 chunk 之间切换清晰度；

实现“自适应流” (Adaptive Streaming) 。

Streaming multimedia: DASH



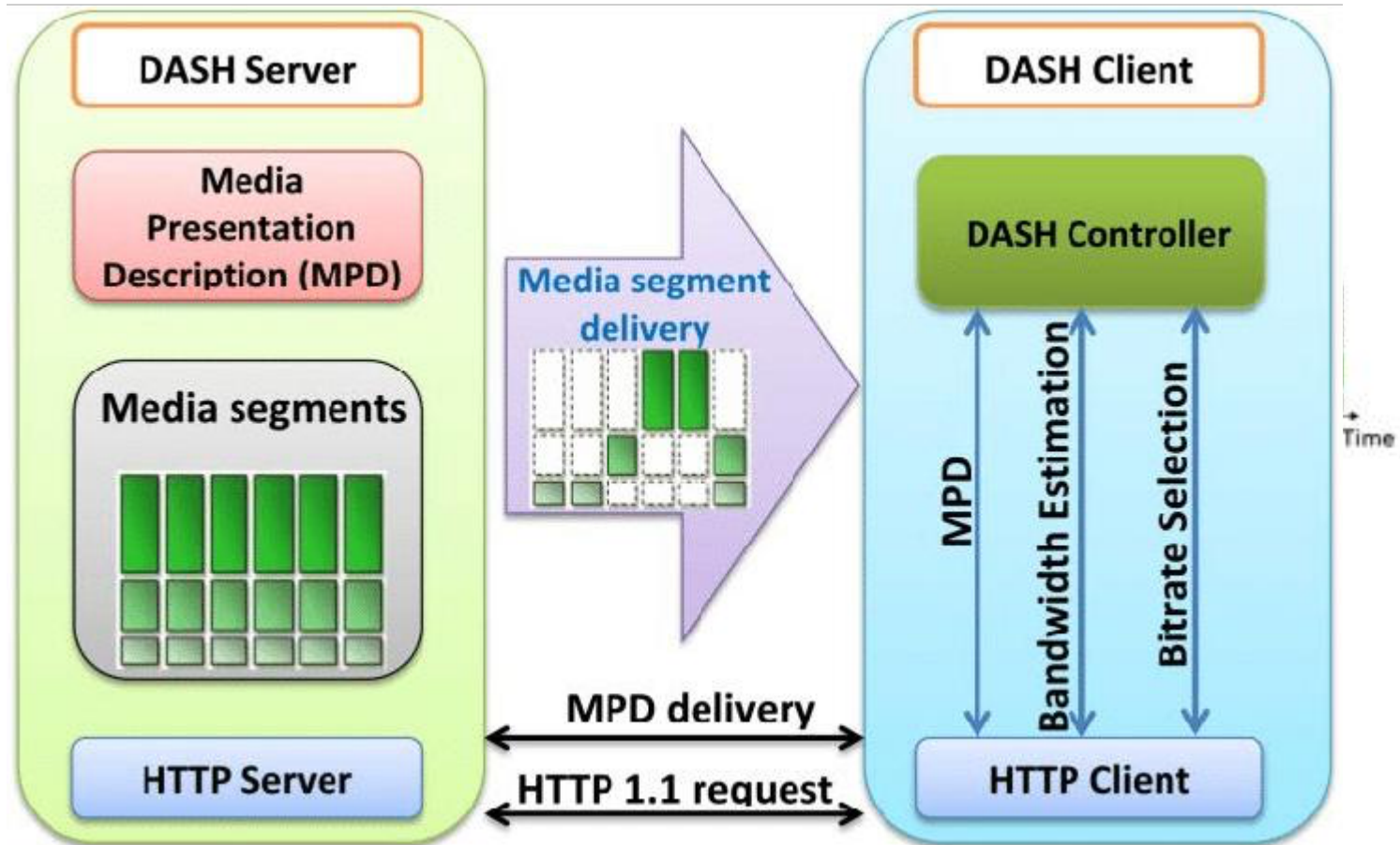
Streaming multimedia: DASH



每个 Representation 里又包括:
Initialization Segment (初始化信息)
Media Segment 1, 2, ... (真正的视频片段)

Media presentation description (MPD), also known as manifest

Streaming multimedia: DASH



Streaming multimedia: DASH

“intelligence” at client: client determines

在合适的时间点请求下一个 chunk;

- **when** to request chunk (so that buffer starvation, or overflow does not occur)
客户端需要控制**缓冲区 (buffer) **的大小:
不能让 buffer 空了 (starvation) → 会导致播放中断;
也不能让 buffer 满了 (overflow) → 浪费带宽和内存
- **what encoding rate** to request (higher quality when more bandwidth available)
- **where** to request chunk (can request from URL server that is “close” to client or has high available bandwidth)

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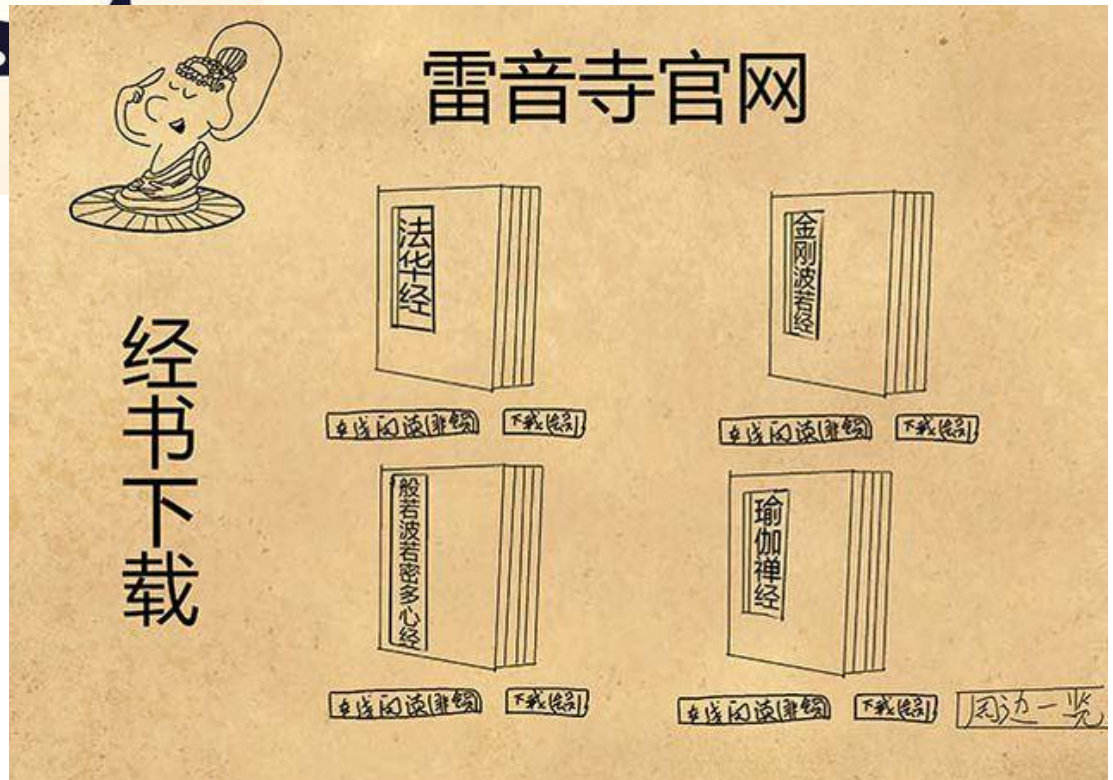
2.6 video streaming and **content distribution networks**

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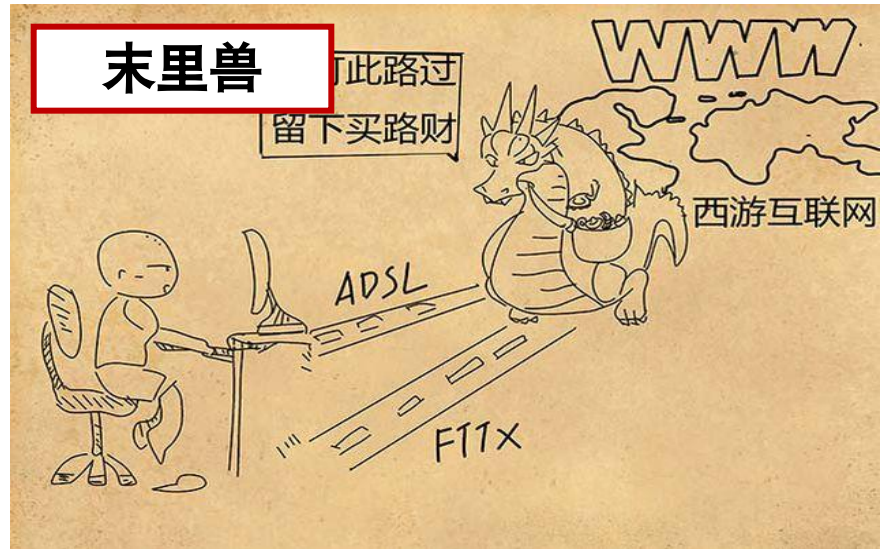
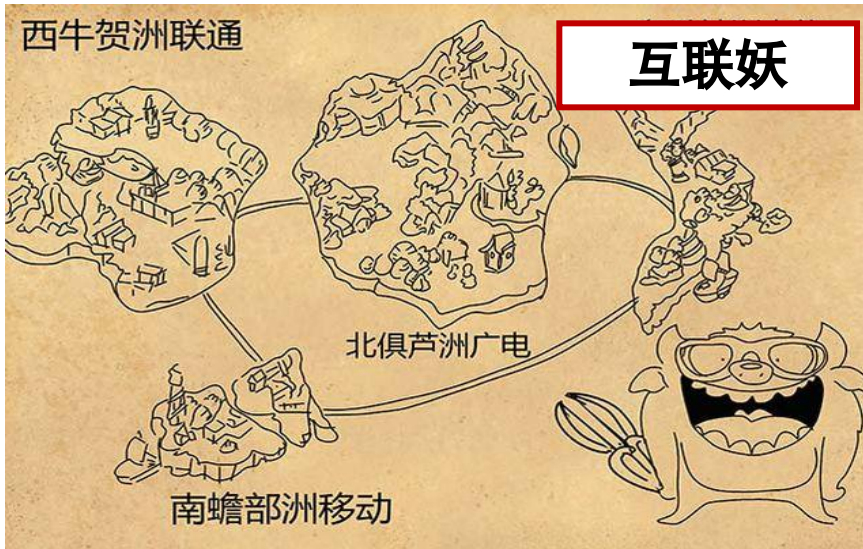
Content distribution networks



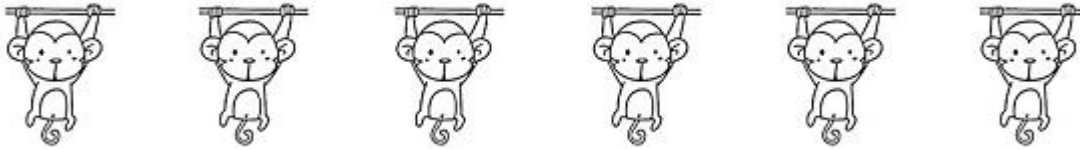
来源: 小黑羊JoinWings
https://baijia.baidu.com/s?old_id=126615&wfr=pc&fr=app_list



Content distribution networks



Content distribution networks



Content distribution networks

- **Challenge:** how to stream content (selected from millions of videos) to hundreds of thousands of simultaneous users?
- **Option 1:** single, large “mega-server”
 - single point of failure
 - huge traffic
 - long path to distant clients
 - multiple copies of video sent over outgoing link

同一个视频可能被传输上百万次；

....quite simply: this solution **doesn't scale**

Content distribution networks

- **Challenge:** how to stream content (selected from millions of videos) to hundreds of thousands of simultaneous users?
- **Option 2: Content distribution networks (CDN)** store/serve multiple copies of videos at multiple geographically distributed sites
 - **Enter deep:** push CDN servers deep into many access networks; inside ISPs
 - close to users
 - used by Akamai, 1700 locations

把 CDN 服务器“深入”部署到各地网络内部，比如放进 ISP（互联网服务提供商）网络。
离用户非常近，延迟最低。
 - **Bring home:** smaller number (10's) of larger clusters in Internet Exchange Point (IXP); outside ISPs
 - used by Limelight

在全球选取少量的大型集群节点；
这些节点部署在 Internet Exchange Points (IXP)，即互联网骨干网的交汇处；
不直接进到 ISP 内部，而是在外部高速连接点上。

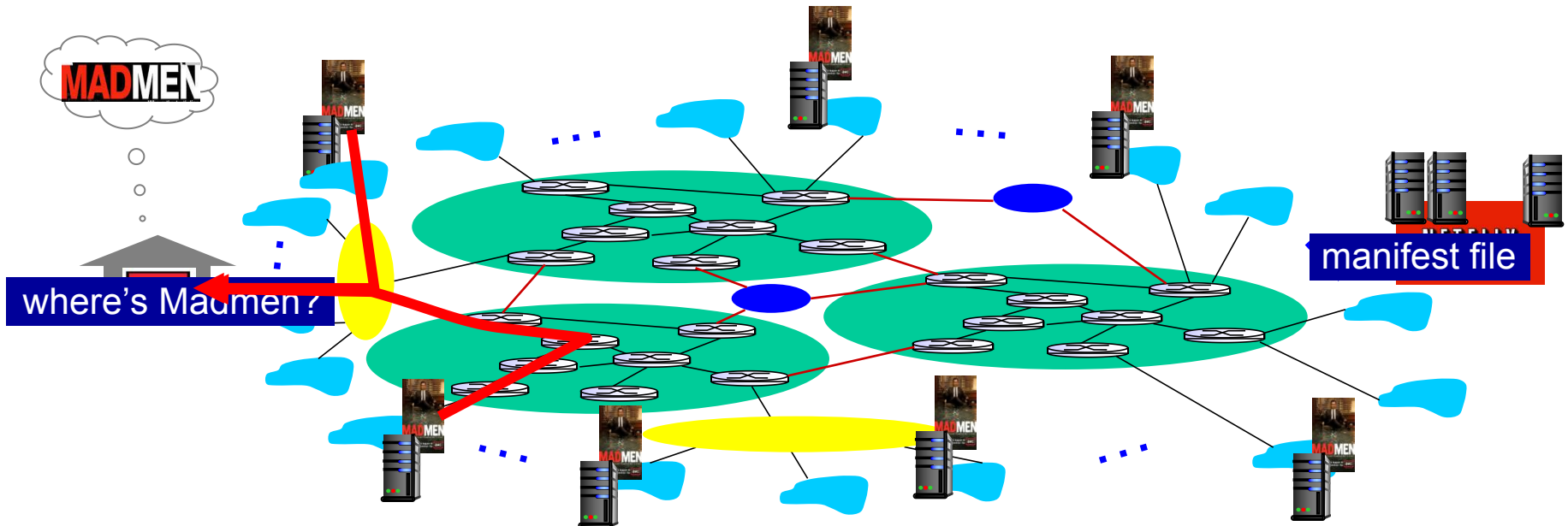
Content Distribution Networks (CDNs)

CDN: stores copies of content at CDN nodes

- e.g. Netflix stores copies of MadMen
- subscriber requests content from CDN
 - directed to nearby copy, retrieves content
 - may choose different copy if network path congested

每个 CDN 节点会保存某些视频或网页的副本；

不同用户实际上访问的是不同服务器上的副本内容，而不是同一个中央服务器。



Content Distribution Networks (CDNs)

Challenges: Coping with a congested Internet

- what content to place in CDN node?

CDN 节点的存储空间有限，不可能保存所有视频。
所以需要决定哪些内容该放进去。

- Simple pull strategy: request, then **store**

- from which CDN node to retrieve content?

- Cluster selection strategy

- the operation for retrieving content?

Simple Pull Strategy (简单拉取策略)

当有用户请求某个内容时，节点从主服务器下载该内容；选择依据包括：

并在本地缓存，以便下次请求时直接提供；

类似浏览器的缓存机制：“request, then store”。

优点：节省空间，只缓存热门内容；
缺点：首次访问时延迟高（因为要去主服务器拉取）。

Cluster Selection Strategy (集群选择策略)。

当同一个内容存在多个节点时，CDN 要决定用户应连接哪个节点。

地理位置（哪个节点离用户近）；

网络延迟/带宽（哪个节点更快）；

负载情况（哪个节点更空闲）；

实时网络状态（避免拥堵路径）。

内容检索与缓存更新策略：

如果请求的内容在本地缓存中：

直接提供给用户；

如果本地没有：

从上级 CDN 节点或主服务器（origin server）
拉取；

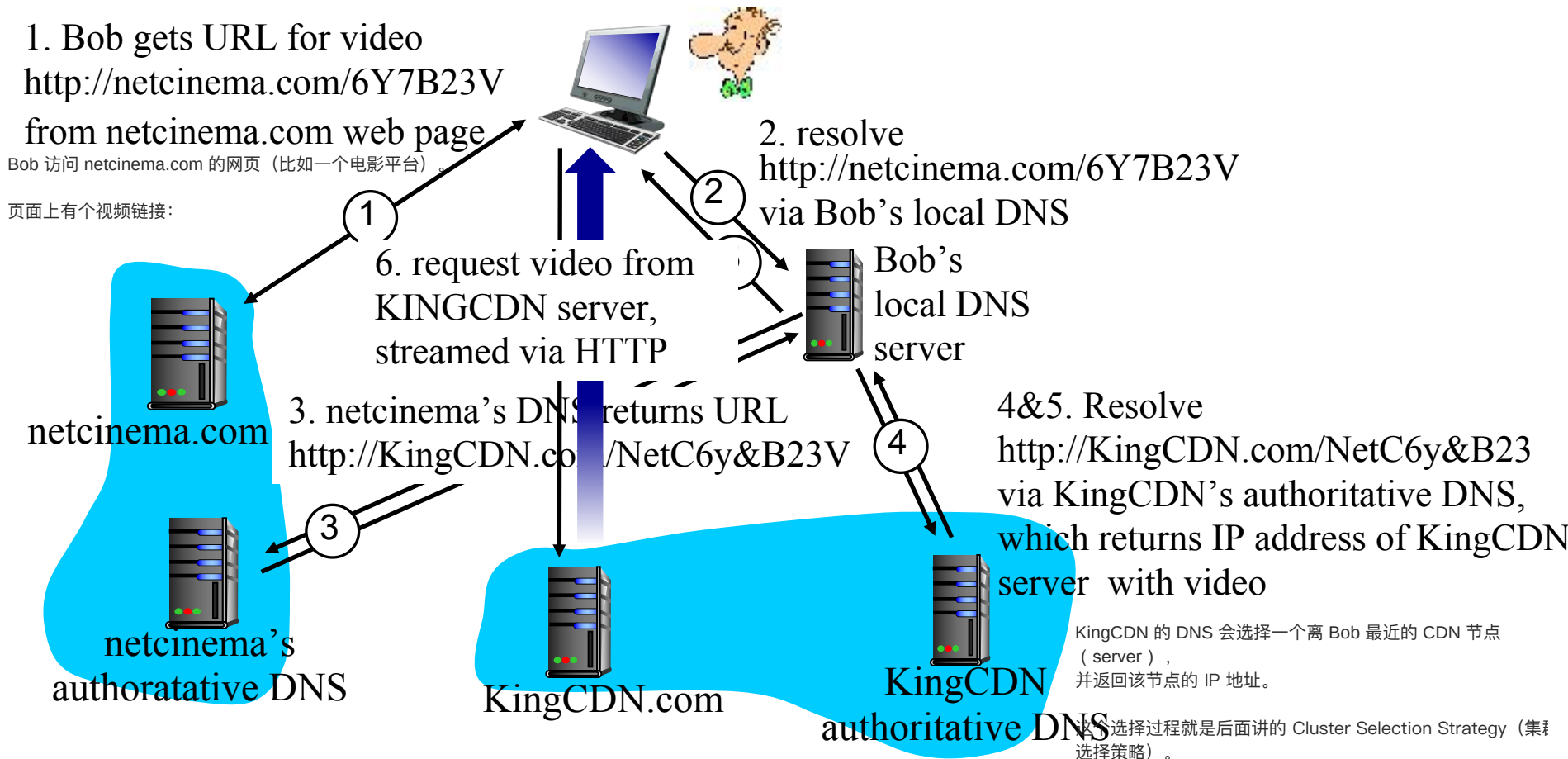
然后缓存一份副本。

这样 CDN 节点之间形成一个分层缓存体系
（hierarchical cache structure）。

CDN Operation

Bob (client) requests video <http://netcinema.com/6Y7B23V>

- video stored in CDN at <http://KingCDN.com/NetC6y&B23V>



CDN: Cluster Selection Strategy

One simple strategy is to assign the client to the cluster that is

geographically closest:

- When a DNS request is received from a particular LDNS, the CDN chooses the geographically closest cluster
- may not be the closest cluster in terms of the length or number of hops
- ignore the variation in delay and available bandwidth over time

Periodic **real-time measurements** of delay and loss performance between their clusters and clients:

- a CDN can have each of its clusters periodically send probes to all of the LDNSs around the world.
- many LDNSs are configured to not respond to such probes.

各 CDN 节点会周期性地向不同地区的本地 DNS 发出探测包 (probe)。

并不是所有的本地 DNS 都会响应这种探测；

记录响应时间，用以动态优化路由。

所以有时候测量结果不完整。

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